# STA 445 S26 Assignment 2

2026-02-24

Load all the packages you will need here. Notice that the code chunk option has been set to include = FALSE. This means the code will run, but the output will not be shown in the knitted document. Thus, you do not need to worry about suppressing warning messages. This code will be helpful when writing a project report.

## Problem 1

Download from GitHub the data file by clicking on this link: Example_5.xls. Open it in Excel and figure out which sheet of data we should import into R. At the same time figure out which range of values to import.

a. Import the dataset into a data frame and show the structure of the imported data using the `str()` command. Make sure that your data has $n = 31$ observations and the three columns are appropriately named. You are not allowed to make any modifications to the data file. All changes must be made within this rmd file.

```r
Example_5 <- read_excel(
  "Example_5.xls",
  sheet = "RawData",
  range = "A6:C36",
  col_names = c("Girth", "Height", "Volume")
)

str(Example_5)
```

```
## tibble [31 x 3] (S3: tbl_df/tbl/data.frame)
##  $ Girth : num [1:31] 8.3 8.6 8.8 10.5 10.7 10.8 11 11 11.1 11.2 ...
##  $ Height: num [1:31] 70 65 63 72 81 83 66 75 80 75 ...
##  $ Volume: num [1:31] 10.3 10.3 10.2 16.4 18.8 19.7 15.6 18.2 22.6 19.9 ...
```

b. Use `slice_sample()` to show five random rows of the dataset.

```r
slice_sample(Example_5, n = 5)
```

```
## # A tibble: 5 x 3
##   Girth Height Volume
##   <dbl>  <dbl>  <dbl>
```

```
## 1  11.4     76   21
## 2  12       75   19.1
## 3   8.8     63   10.2
## 4  13.8     64   24.9
## 5   8.3     70   10.3
```

## Problem 2

Download from GitHub the data file by clicking on this link: Example_3.xls. Import the data set into a data frame and show the structure of the imported data using the `str()` command. Make sure the Tesla values are `NA` where appropriate and that both `-9999` and `NA` are imported as NA values. You are not allowed to make any modifications to the data file. All changes must be made within this rmd file.

```
Example_3 <- read_excel("Example_3.xls")
```

```
## New names:
## * `` -> `...2`
```

b. Use the `tail()` command to show the last five rows of a data table.

```
tail(Example_3, 5)
```

```
## # A tibble: 5 x 2
##   The data is from the R data frame 'mtcars' and I simply copy and paste~1 ...2
##   <chr>                                                                    <chr>
## 1 gear                                                                     Numb~
## 2 carb                                                                     Numb~
## 3 <NA>                                                                     <NA>
## 4 <NA>                                                                     <NA>
## 5 I added a 2020 Tesla model S.  I used NA to denote values that aren't a~ <NA>
## # i abbreviated name:
## #   1: `The data is from the R data frame 'mtcars' and I simply copy and pasted into Excel. Th
```

c. Use `slice_sample()` to show five random rows of the dataset.

```
slice_sample(Example_3, n = 5)
```

```
## # A tibble: 5 x 2
##   The data is from the R data frame 'mtcars' and I simply copy and paste~1 ...2
##   <chr>                                                                    <chr>
## 1 hp                                                                       Gros~
## 2 wt                                                                       Weig~
## 3 gear                                                                     Numb~
## 4 The data columns are:                                                    <NA>
## 5 carb                                                                     Numb~
## # i abbreviated name:
## #   1: `The data is from the R data frame 'mtcars' and I simply copy and pasted into Excel. Th
```

## Problem 3

Download all of the files from GitHub `data-raw/InsectSurveys` directory which have been linked here. Each month's file contains a sheet with information about each of the sites that was surveyed. The second sheet contains information about the number of each species that was observed at each site.

a. Import the data for each month and create a single `site` data frame with information from each month. You are not allowed to make any modifications to the data file. All changes must be made within this rmd file. You may need to change column names and reorganize columns.

```r
read_site_month <- function(file, month) {
  read_excel(file, sheet = 1) %>%
    filter(!if_all(everything(), is.na)) %>%
    mutate(Month = month)
}

May_site  <- read_site_month("~/Documents/sta_445/May.xlsx", "May")
June_site <- read_site_month("~/Documents/sta_445/June.xlsx", "June")
July_site <- read_site_month("~/Documents/sta_445/July.xlsx", "July")
Aug_site  <- read_site_month("~/Documents/sta_445/August.xlsx", "August")
Sep_site  <- read_site_month("~/Documents/sta_445/September.xlsx", "September")
Oct_site  <- read_site_month("~/Documents/sta_445/October.xlsx", "October")

site <- bind_rows(
  May_site,
  June_site,
  July_site,
  Aug_site,
  Sep_site,
  Oct_site
)
```

b. Show the structure of the data frame using the `str()` command.

```r
str(site)
```

c. Use `slice_sample()` to show eight random rows of the dataset.

```r
slice_sample(site, n = 8)
```

d. Comment on the importance of consistency of your data input sheets.

A standard data input sheet effectively makes it easy to combine data from multiple files without error For example, using standard column names, data types, units, and formats makes it easy to use functions such as bind_rows() and automate analyzes Or if there are other issues with the inputs, such as missing values, misalignment or import errors, the analysis becomes more error-prone and less reproducible